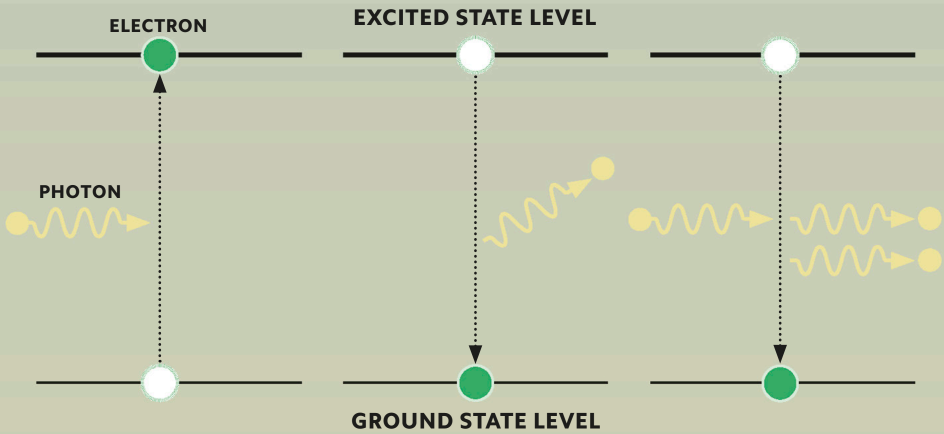
FIRING PHOTONS AT ATOMS

When an electron falls from an excited state to its ground state (see p.30), it releases the energy difference as a photon. Stimulated emission involves firing photons at atoms, causing excited electrons to fall to their ground states and release new photons with wavelength and phase identical to the incident photons. These stimulate other atoms, resulting in a cascade of coherent light (see opposite).



Absorption of light
When an electron absorbs energy from an incident photon, it may be excited (also referred to as being "pumped") to a higher energy state. An electron in an excited state may then decay to a lower energy state.

Spontaneous emission of light
When an electron falls to a lower state spontaneously, it releases energy in the form of a photon. The phase and direction of photons spontaneously emitted by electrons in a material is random.

Stimulated emission of light
An incident photon can cause an electron to fall to a lower energy state. In the process, the electron emits an additional photon, which has the same phase and direction as the incident photon.